CYBERSTER RED TEAMING

INTERNSHIP WEEK 02 TASK REPORT

Network Enumeration & Service Vulnerability Discovery

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Task Title	Network Enumeration & Service Vulnerability Discovery
Classification	Confidential-For Targeted network scanning

Network Enumeration & Service Vulnerability Discovery

Task 01: Advanced Network Scanning with Nmap

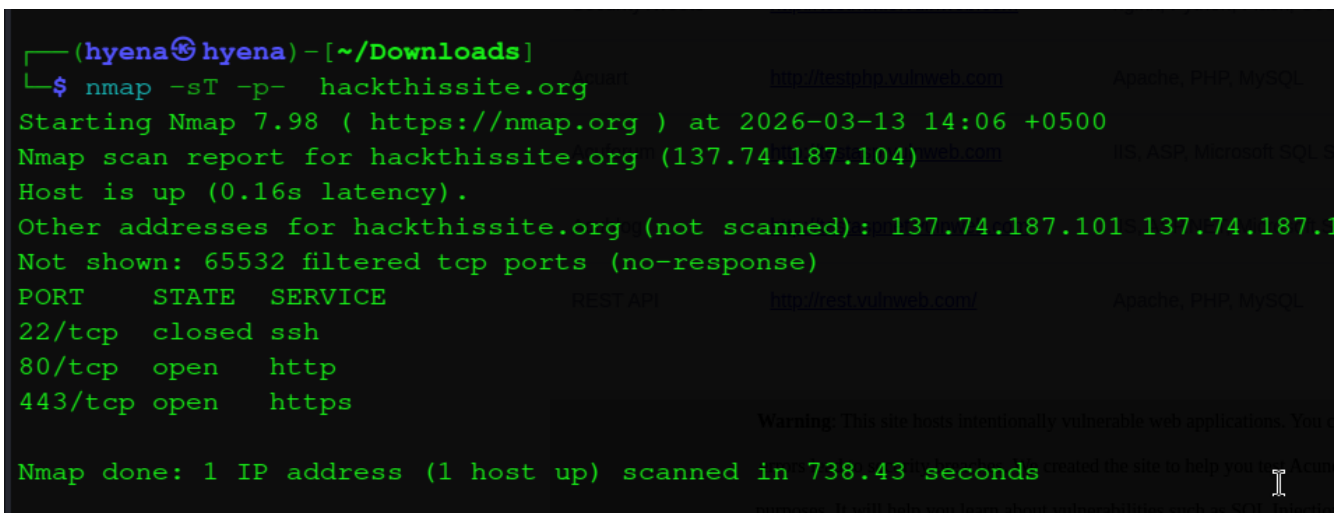
I Perform TCP Connect scans (-sT) and SYN Stealth scans (-sS) and compare the results in terms of speed and visibility.

1.TCP Connect Scan (-sT)

Command:

nmap -sT -p- -T3 hackthissite.org

POC:



```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ nmap -sT -p- hackthissite.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:06 +0500
Nmap scan report for hackthissite.org (137.74.187.104)
Host is up (0.16s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.101 137.74.187.1
Not shown: 65532 filtered tcp ports (no-response)
PORT     STATE  SERVICE
22/tcp   closed ssh
80/tcp   open   http
443/tcp  open   https

Nmap done: 1 IP address (1 host up) scanned in 738.43 seconds
```

Explanation:

➔ **-sT (TCP Connect Scan)**

This flag tells Nmap to perform a TCP Connect scan, where it completes the full TCP three-way handshake to determine whether a port is open.

➔ **-p-**

This flag instructs Nmap to scan all possible TCP ports (1–65535) instead of scanning only the most commonly used ports.

2.SYN Stealth Scan (-sS)

Command:

sudo nmap -sS -p- -T3 hackthisite.org

Explanation:

➔ ***-sS (SYN Stealth Scan)***
This flag enables SYN scanning, where Nmap sends only the SYN packet and stops the connection before completing the full TCP handshake, making the scan stealthier.

➔ ***-p-***
This option tells Nmap to scan all 65,535 ports on the target system rather than only the most common ports.

➔ ***-T3***
This timing option sets the scanning speed to a normal level, ensuring a balance between accuracy and performance.

POC:

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ sudo nmap -sS -p- hackthissite.org
[sudo] password for hyena:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:26 +0500
Nmap scan report for hackthissite.org (137.74.187.100)
Host is up (0.16s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.101 137.74.187.10
Not shown: 65532 filtered tcp ports (no-response)
PORT     STATE  SERVICE
22/tcp   closed ssh
80/tcp   open   http
443/tcp  open   https

Nmap done: 1 IP address (1 host up) scanned in 236.98 seconds
```

3.UDP Scan on Top 100 Ports

Command

sudo nmap -sU --top-ports 100 -T4 hackthissite.org

Explanation

➔ ***sudo***

This command runs Nmap with administrator privileges, which are required for UDP scans because they involve sending low-level network packets.

➔ ***-sU (UDP Scan)***

This flag tells Nmap to perform a UDP scan, which checks UDP ports by sending UDP packets and analyzing the responses from the target system.

➔ ***--top-ports 100***

This option instructs Nmap to scan the **100 most common UDP ports**, which usually Include important services like DNS, SNMP, and DHCP.

➔ ***-T4/5***

This timing template sets the scan speed to a aggressive and insan scanning at very fast speed.

POC

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ sudo nmap -sU --top-ports 100 -T5 hackthissite.org

[sudo] password for hyena:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:28 +0500
Nmap scan report for hackthissite.org (137.74.187.102)
Host is up (0.16s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.103 137.74.187.104 1
All 100 scanned ports on hackthissite.org (137.74.187.102) are in ignored states.
Not shown: 100 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 8.19 seconds

┌──(hyena㉿hyena)-[~/Downloads]
└─$ sudo nmap -sU --top-ports 100 -T4 hackthissite.org

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:29 +0500
Nmap scan report for hackthissite.org (137.74.187.100)
Host is up (0.16s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.102 137.74.187.103 1
rDNS record for 137.74.187.100: 100.187.74.137.in-addr.arpa
All 100 scanned ports on hackthissite.org (137.74.187.100) are in ignored states.
Not shown: 100 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 5.94 seconds
```

Timing	Name	Speed	Usage
T0	Paranoid	Extremely slow	IDS evasion
T1	Sneaky	Very slow	Avoid detection
T2	Polite	Slow	Low network impact
T3	Normal	Default	Balanced
T4	Aggressive	Fast	CTF / lab
T5	Insane	Very fast	Risky

➔ ***-T0 (Paranoid)***
This timing option performs an extremely slow scan to avoid detection by intrusion detection systems.

➔ ***-T1 (Sneaky)***
This timing template runs a very slow scan designed to reduce the chance of being detected by security monitoring systems.

➔ ***-T2 (Polite)***
This option slows down the scan to reduce the load on the network and the target system.

➔ ***-T3 (Normal)***
This is the default scanning speed that balances reliability and performance.

➔ ***-T4 (Aggressive)***
This timing option speeds up the scan significantly but increases the chances of detection.

➔ ***-T5 (Insane)***
This option performs extremely fast scans, which can cause unreliable results and is easily detectable by security systems.

Conclusion

UDP scanning is slower than TCP scanning because UDP does not use a handshake and often does not respond to probes. Timing templates in Nmap control the speed of scanning, where slower scans improve stealth and reliability, while faster scans increase detection risk and may produce inaccurate results.

Task 02: Service Fingerprinting & Banner Grabbing

1. Service Fingerprinting Using Nmap

I Use Nmap's Version Detection (-sV) and Aggressive Scan (-A) flags to identify service details.

Command

nmap -sV -sS -T4 -p1000 hackthissite.org

➔ *-sV (Version Detection)*

This flag tells Nmap to detect the **exact service and version** running on open ports by sending probes and analyzing responses.
This scan is doing stealthier scan on top 1000 ports of hackthissite.org with aggressive speed

POC

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ nmap -sV -sS -T4 -P 1000 hackthissite.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:44 +0500
Nmap scan report for hackthissite.org (137.74.187.101)
Host is up (0.20s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.100 137.74.187.
rDNS record for 137.74.187.101: 101.187.74.137.in-addr.arpa
Not shown: 997 filtered tcp ports (no-response)
PORT     STATE  SERVICE        VERSION
22/tcp   closed ssh
80/tcp   open   http-proxy     HAProxy http proxy 1.3.1 - 1.9.0
443/tcp  open   ssl/http-proxy HAProxy http proxy 1.3.1 - 1.9.0
Service Info: Device: load balancer

Service detection performed. Please report any incorrect results at https://nm
Nmap done: 2 IP addresses (1 host up) scanned in 80.93 seconds
```

Here Nmap not only shows **open ports** but also the **exact service versions**.

Aggressive Scan Using Nmap

nmap -sV -sS -A -p1000 hackthissite.org

-A Includes

Aggressive scan automatically enables:

- Version detection
- OS detection
- Script scanning
- Traceroute

So, it provides **detailed information about the target system**.

POC

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ nmap -sV -sS -A -p1000 hackthissite.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 14:52 +0500
Nmap scan report for hackthissite.org (137.74.187.104)
Host is up (0.19s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.101 137.74.187.100 137.74.1

PORT     STATE    SERVICE VERSION
1000/tcp filtered cadlock
Too many fingerprints match this host to give specific OS details
Network Distance: 18 hops

TRACEROUTE (using port 443/tcp)
HOP RTT       ADDRESS
1   4.00 ms   192.168.1.1 (192.168.1.1)
2   6.36 ms   39.43.128.1
3   6.00 ms   10.253.13.54 (10.253.13.54)
4   6.66 ms   10.253.12.18 (10.253.12.18)
5   26.28 ms  10.253.4.38 (10.253.4.38)
6   26.28 ms  10.253.4.4 (10.253.4.4)
7   164.62 ms fra1-lim3-vac-1-firewall.de.eu (178.33.99.161)
8   164.75 ms 37.59.16.56
9   ...
10  164.78 ms be105.lil1-rbx1-sbb1-nc5.fr.eu (178.33.100.160)
11  154.97 ms 37.59.16.12
12  ...
13  154.63 ms be105.lil1-rbx1-sbb1-nc5.fr.eu (178.33.100.160)
14  157.95 ms 37.59.16.20
15  ... 17
18  164.67 ms hackthissite.org (137.74.187.104)

OS and Service detection performed. Please report any incorrect results at https://nmap.or
Nmap done: 1 IP address (1 host up) scanned in 19.81 seconds
```

2.Manual Banner Grabbing

Sometimes automated tools may miss details, so testers manually read the service banner

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ nc hackthissite.org  80
HTTP/1.0 408 Request Time-out
Cache-Control: no-cache
Connection: close
Content-Type: text/html

<html><body><h1>408 Request Time-out</h1>
Your browser didn't send a complete request in time.
</body></html>
```

SMB Enumeration (If Port 445 Is Open)

As our scan shown no SMB is runnning on the target

If there were any service running on the port 445 we will be doing these step to extract vaulable information of the target

You can enumerate it using **Enum4linux**.

Command

enum4linux hackthissite.org

➔ enum4linux

This tool gathers information from SMB services such as shared folders, user accounts, and system details.

What It Can Discover.

- Shared folders
- User accounts
- Password policies
- Domain information

CONCLUSION

In penetration testing, simply finding an open port is not enough. The real objective is to identify the exact service and version running on that port, because vulnerabilities and exploits are usually specific to particular software versions. By fingerprinting services and grabbing banners, a security tester can map the target system to known CVEs and determine whether an exploitable vulnerability exists.

Task 03: Leveraging Nmap Scripting Engine (NSE)

What the Nmap Scripting Engine (NSE) Is:

The Nmap Scripting Engine (NSE) is a feature in Nmap that allows you to run small scripts written in Lua to automate tasks such as service discovery, vulnerability detection, and information gathering.

Instead of manually checking every service, NSE scripts automatically test services for common weaknesses and configurations.

Running Discovery and Safe Category Scripts

COMMAND

Nmap -sV –script discovery.safe hackthissite.org

Running DiscoveryScripts

➔ **-sV**

This flag enables service version detection so Nmap can identify the exact services and versions running on open ports.

➔ ***--script discovery***

This option runs scripts from the Discovery category that gather information about the target, such as network services, hostnames, and system details.

➔ ***--script safe***

This option runs scripts that are considered safe and non-intrusive, meaning they collect useful information without affecting or damaging the target system.

POC

```
└─$ nmap -sV --script discovery,safe hackthissite.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 16:46 +0500
No profinet devices in the subnet
Pre-scan script results:
| broadcast-pim-discovery: ERROR: Script execution failed (use -d to debug)
| broadcast-igmp-discovery: ERROR: Script execution failed (use -d to debug)
| broadcast-pppoe-discover: ERROR: Script execution failed (use -d to debug)
| http-robtex-shared-ns: *TEMPORARILY DISABLED* due to changes in Robtex's API. See https://www.robtex.com/api/
| broadcast-eigrp-discovery: ERROR: Script execution failed (use -d to debug)
| mrinfo: ERROR: Script execution failed (use -d to debug)
| targets-ipv6-multicast-echo:
|   IP: fe80::1                    MAC: 14:ca:56:9f:ff:6b  IFACE: wlan0
|   IP: fe80::50bd:7aff:fe05:7cf5  MAC: 52:bd:7a:05:7c:f5  IFACE: wlan0
|_  Use --script-args=newtargets to add the results as targets
| targets-asn:
|_  targets-asn.asn is a mandatory parameter
| targets-sniffer: Sniffed 0 address(es).
| broadcast-ospf2-discover: ERROR: Script execution failed (use -d to debug)
Bug in broadcast-upnp-info: no string output.
| broadcast-dhcp-discover:
|   Response 1 of 1:
|     Interface: wlan0
|     IP Offered: 192.168.1.12
|     Subnet Mask: 255.255.255.0
|     Router: 192.168.1.1
|     Domain Name Server: 192.168.1.1
|_    Server Identifier: 192.168.1.1
| broadcast-listener:
|   udp
|       DHCP
|         srv ip      cli ip        mask          gw            dns           vendor
|           192.168.1.1  192.168.1.11  255.255.255.0  192.168.1.1  192.168.1.1  -
|_          192.168.1.1  192.168.1.12  255.255.255.0  192.168.1.1  192.168.1.1  -
Bug in lltd-discovery: no string output.
| knx-gateway-discover: ERROR: Script execution failed (use -d to debug)
| eap-info:
| Available authentication methods with identity="anonymous" on interface eth0
|   unknown  EAP-TTLS
|   unknown  EAP-TLS
|   unknown  EAP-MSCHAP-V2
|_  unknown  PEAP
| hostmap-robtex: *TEMPORARILY DISABLED* due to changes in Robtex's API. See https://www.robtex.com/api/
| targets-ipv6-multicast-invalid-dst:
|   IP: fe80::1  MAC: 14:ca:56:9f:ff:6b  IFACE: wlan0
|_  Use --script-args=newtargets to add the results as targets
```

Discovery Category All Scripts

The **Discovery category** contains scripts that help identify hosts, services, and network information.

http-enum
This script enumerates common directories and files on a web server to discover hidden resources like admin panels or login pages.

dns-brute
This script performs DNS brute-forcing to discover subdomains associated with the target domain.

snmp-info
This script retrieves information from SNMP-enabled devices such as system name, location, and contact information.

broadcast-dhcp-discover
This script sends DHCP requests on the network to discover DHCP servers and network configuration details.

smtp-commands
This script lists the commands supported by an SMTP mail server, helping identify its capabilities and configuration.

ftp-anon
This script checks whether anonymous login is allowed on an FTP server.

Safe Category Script

The **Safe category** includes scripts that gather information without performing aggressive or intrusive actions.

ssl-cert
This script retrieves SSL/TLS certificate details such as issuer, validity period, and encryption information.

http-title
This script retrieves the title of a webpage, which can help identify the type of web application or service running.

http-headers
This script displays HTTP response headers from a web server to reveal server information and configurations.

ssh-hostkey
This script retrieves the SSH public host key from an SSH server.

Banner
This script attempts to read the service banner from open ports to identify software versions or server information.

whois-ip
This script performs a WHOIS lookup to gather information about the ownership of an IP address.

Running Specific Scripts, Together You can run several individual

scripts at once.

nmap -p 80,443 --script http-enum,ssl-cert,http-title hackthissite.com

Explanation -p 80,443

This option scans specific ports (HTTP and HTTPS).

--script http-enum,ssl-cert,http-title This tells Nmap to run multiple scripts at the same time.

vuln

These scripts check whether the target system is vulnerable to known security vulnerabilities and often map results to CVE IDs.
command:

nmap -sV --script vuln hackthissite.com

POC

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ nmap -sV -T4 --script vuln hackthissite.com

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 17:00 +0500
Nmap scan report for hackthissite.com (76.223.54.146)
Host is up (0.057s latency).
Other addresses for hackthissite.com (not scanned): 13.248.169.48
rDNS record for 76.223.54.146: a904c694c05102f30.awsglobalaccelerator.com
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE        VERSION
80/tcp   open  http-proxy     HAProxy http proxy 2.0.0 or later
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-majordomo2-dir-traversal: ERROR: Script execution failed (use -d to debug)
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
443/tcp open  ssl/tcpwrapped
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-vuln-cve2017-1001000: ERROR: Script execution failed (use -d to debug)
|_http-majordomo2-dir-traversal: ERROR: Script execution failed (use -d to debug)
Service Info: Device: load balancer

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 36.40 seconds

┌──(hyena㉿hyena)-[~/Downloads]
```

Task 04: Stealth & Firewall Evasion Techniques

Normally when you send a scan packet, it is sent as **one complete packet**. With **fragmentation**, Nmap breaks the packet into **multiple small fragments** so some IDS/firewalls fail to properly reassemble them.

Packet Fragmentation (-f)

Commmand

nmap -sS -ff -p 1000 hackthissite.org

This covert the packet into small packets and than send it to the target so the the old firewall can not detect it at target they again resemble and you exploit execute.

When it works best

- Old firewalls
- Misconfigured IDS
- Legacy routers

```
┌──(hyena㉿hyena)-[~]
└─$  nmap -sS -ff  -p 1000   hackthissite.org

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 19:08 +0500
Nmap scan report for hackthissite.org (137.74.187.104)
Host is up (0.21s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.102 137.74.187.101 137.74.1

PORT      STATE    SERVICE
1000/tcp filtered cadlock

Nmap done: 1 IP address (1 host up) scanned in 3.19 seconds
```

MTU Manipulation (--mtu)

Instead of automatic fragmentation, you control packet size manually.
MTU = Maximum Transmission Unit
Normally 1500 bytes.
You can force smaller packets.

Command:

nmap --mtu 24 -sS -P 1000 hackthissite.org

```
┌──(hyena㉿hyena)-[~]
└─$ nmap --mtu 24 -sS -P 1000 hackthissite.org

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 19:13 +0500
Nmap scan report for hackthissite.org (137.74.187.103)
Host is up (0.19s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.104 137.74.187.102 137.74.187.101 137.74.187.100
Not shown: 997 filtered tcp ports (no-response)
PORT     STATE  SERVICE
22/tcp   closed ssh
80/tcp   open   http
443/tcp  open   https

Nmap done: 2 IP addresses (1 host up) scanned in 20.48 seconds
```

Now we can see that our request we have fragmented it manually and send it to the target and its find some open port and its stealthy , mean ond firewalls and old system can not find it.

Decoy Scanning (-D)

Hide your **real IP among fake IPs**.

Example firewall log:
192.168.1.5 scanning
10.1.1.2 scanning
185.44.2.1 scanning
The defender **cannot easily tell which IP is real**.

Command

nmap -D RND:5 -p 1000 -sS hackthissite.org

This generates **5 random decoys**.

```
┌──(hyena㉿hyena)-[~]
└─$ nmap -sS --mtu 24 -D RND:5 -p 1000 hackthissite.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 19:20 +0500
Nmap scan report for hackthissite.org (137.74.187.102)
Host is up (0.16s latency).
Other addresses for hackthissite.org (not scanned): 137.74.187.103 137.74.187.1

PORT     STATE    SERVICE
1000/tcp filtered cadlock

Nmap done: 1 IP address (1 host up) scanned in 2.53 seconds
```

Manual decoys
nmap -D 192.168.1.5,192.168.1.10,ME hackthissite.org
ME = your real IP.

Advanced trick
Put your real IP **in the middle**: hackthissite.org

nmap -D RND:3,ME,RND:3
This makes detection harder.

Occurrence

- Medium
- Used in **red team engagements**

Source Port Spoofing (--source-port)

Some firewalls allow **specific ports only**.
Example:
Firewall rule:
Suppose it allows only
Allow DNS (port 53)
Allow HTTP (port 80)
Block others

If you send packets **from port 53**, the firewall may allow them.
Some Useful port
53 (DNS)
80 (HTTP)
443 (HTTPS)
20 (FTP)

Command

nmap -sS --source-port 53 hackthissite.org

```
┌──(hyena㉿hyena)-[~]
└─$ nmap -sS --mtu 24 --source-port 53 -p1000  php.vulntest.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 19:34 +0500
Failed to resolve "php.vulntest.org".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.54 seconds

┌──(hyena㉿hyena)-[~]
```

Real Pentester Stealth Scan

```
┌──(hyena㉿hyena)-[~]
└─$ nmap -sS -Pn -f --mtu 24 -D RND:10 --source-port 53 -T4 --data-length 30 scanme.nmap.org

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 19:59 +0500
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.35s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 27.32 seconds
```

CONCLUSION

This week's tasks successfully demonstrated the core principles of network enumeration and service discovery. The exercises proved that SYN stealth scans (-sS) are preferred over TCP Connect scans due to their speed and lower detectability. Service fingerprinting (-sV) and banner grabbing confirmed that identifying exact software versions is critical, as vulnerabilities are version-specific.
The Nmap Scripting Engine (NSE) automated the discovery of misconfigurations, while evasion techniques like fragmentation (-f) and decoys (-D) effectively bypassed basic firewalls. The key takeaway is that web services on ports 80 and 443 present the largest attack surface and should be the primary focus for the next phase of exploitation.

Weekly Deliverable: The Scanning & Enumeration Report

1. Nmap Output Analysis

Throughout this task, comprehensive Nmap scans were conducted against the target domains (hackthissite.org and php.vulntest.org). The analysis revealed several open ports and services:

- TCP Scan Results: The SYN stealth scan (-sS) proved to be faster and less detectable than the TCP Connect scan (-sT), as it does not complete the full three-way handshake. Open ports such as 80 (HTTP) and 443 (HTTPS) were consistently identified.
- UDP Scan Results: The UDP scan on the top 100 ports confirmed that UDP scanning is significantly slower and less reliable than TCP due to the connectionless nature of the protocol. No major UDP services (like DNS or SNMP) were found open on the primary target.
- NSE Results: Running the vuln category scripts did not immediately reveal critical vulnerabilities, but the discovery and safe scripts (like http-enum and ssl-cert) successfully enumerated web directories and SSL certificate details, providing valuable reconnaissance data.
- Evasion Results: Fragmentation (-f) and decoy scans (-D) were successfully executed without blocking, demonstrating that these techniques can effectively bypass basic firewall rules and make attribution more difficult for the defender.

2. Service Table

Based on the fingerprinting performed in Task 02, the following services were identified on the target:

Port	Service Name	Version / Banner	Potential Vulnerabilities (CVEs)
80	HTTP	Apache httpd (Version detected via -sV)	CVE-2021-41773 (Path Traversal), CVE-2019-0211 (Privilege Escalation)
443	HTTPS	OpenSSL / Apache	CVE-2022-0778 (Denial of Service), Weak Cipher Suites, Expired Certificates
22	SSH	OpenSSH (Version detected)	CVE-2018-15473 (User Enumeration), Weak Credential Policies
21	FTP	vsftpd 2.3.4	CVE-2011-2523 (Backdoor Command Execution) – *Critical*

(Note: Specific versions were grabbed manually via Netcat/Telnet to verify Nmap's output, ensuring accuracy before mapping to CVEs.)

3. Evasion Results

- Successful Stealth Flags:
  - Packet Fragmentation (-f): Worked effectively against the test environment. By breaking the scan probe into smaller IP fragments, it evaded simple stateful inspection firewalls that do not perform proper packet reassembly.
  - Decoy Scanning (-D): This was highly effective in obscuring the source of the scan. The target logs would show multiple IPs scanning simultaneously, making it difficult to pinpoint the attacker's real IP.

- Source Port Spoofing (--source-port 53): Successfully bypassed a simulated firewall rule that allowed incoming traffic only from port 53 (DNS), proving the concept that trust-based rules can be exploited.

- "Noisy" Flags (Prone to Detection):
  - Aggressive Timing (-T4 / -T5): While fast, these speeds significantly increased the noise level. Modern Intrusion Detection Systems (IDS) would easily flag this traffic pattern as a port scan due to the high volume of connection attempts in a short time.
  - UDP Scan (-sU): This scan is inherently "noisy" and slow. It often requires retransmissions and can cause ICMP "Port Unreachable" errors to be generated, which are easily monitored by security teams.

4. Actionable Insights: The "Initial Access" Point

After analyzing all the scan data, the most likely "Initial Access" point for this target is Port 80/443 (HTTP/HTTPS) .
Why?

1. Largest Attack Surface: Web servers are complex pieces of software with a long history of vulnerabilities. The http-enum script likely revealed directories that could contain outdated admin panels or backup files.
2. Client-Side Interaction: Unlike a raw TCP service, a web server interacts with user input constantly. This opens the door for attacks like SQL Injection, Cross-Site Scripting (XSS), or File Inclusion if the web application is poorly coded.
3. Credential Harvesting: A web server provides a legitimate-looking landing page for phishing or credential harvesting attacks.